

Homoplantagin Antagonizes *N*-Methyl-D-aspartate Receptor and Extracellular Signal-Regulated Kinase Signaling in $A\beta$ Oligomers-Induced Neuropathology/Toxicity

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ABSTRACT: Extracts from plants/herbals are great resources of drugs and nutrients. Baicalein, a component present in *Scutellaria baicalensis*, was previously found to alleviate the abnormal depolarization brought about by $A\beta$ oligomers. We extended this promising outcome by screening baicalein derivatives, and a natural compound named homoplantagin was pinpointed. In this study, we aimed to investigate the effects of homoplantagin on animal behavior and explore its neuronal functioning/mechanism. In behavior tests, impairments of novel object recognition and of spatial learning/memory were reversed by homoplantagin in a J20 Alzheimer's disease (AD) mouse model. Utilizing primary glutamatergic neurons, homoplantagin was found to prevent the $A\beta$ oligomer-induced increase in ERK phosphorylation. Furthermore, homoplantagin inhibits both AMPA-insult and NMDA-insult depolarization; this was assessed using DiBAC4(3), a membrane potential sensitive dye. Finally, homoplantagin blocks both $A\beta$ oligomer-induced and NMDA-induced calcium influx, which was assessed by intracellular calcium measurement using Fura2/AM. Interestingly, homoplantagin immediately blunts the steady state calcium influx caused by NMDA. Taken together, homoplantagin is capable of inhibiting $A\beta$ oligomer-induced pathophysiology, in particular at the receptor level. This pure compound has great potential to be developed as a clinical therapeutic drug.

KEYWORDS: Alzheimer's disease, behavior tests, calcium influx, depolarization, ERK, homoplantagin

INTRODUCTION

Plant extracts have been found to have great merit as a source of clinical drugs, and examples include metformin for diabetes and taxol for ovarian cancer, as well as many others. Alzheimer's disease (AD) is the most common dementia among the elderly. Only four small molecule drugs, namely the acetylcholinesterase inhibitors donepezil, rivastigmine, and galantamine, together with the NMDAR antagonist memantine, are on the current U.S. Food and Drug Administration (FDA) approved list of clinical drugs used for the treatment of AD patients.^{1,2} In addition, two antibody drugs, aducanumab and lecanemab, which bind with $A\beta$ oligomers and/or $A\beta$ fibrils in order to prevent the formation of $A\beta$ plaques and accelerate the clearance of these complexes, have been recently approved by the U.S. FDA.³ The majority of molecules tested up to the present in this context are synthesized compounds. Obviously, there is an urgent need to develop more drugs to treat AD and thus there is a need to target as wide a range of sources and molecular mechanisms as possible. In particular, drugs from natural sources are an important resource.

Currently, beta amyloid 1–42 ($A\beta$ 1–42) oligomers/aggregation and tau hyperphosphorylation are the best-known hallmarks of AD. The increase in the abnormal $A\beta$ configuration and the hyperphosphorylation of tau protein are both involved in bringing about neuronal death.^{4–6} During $A\beta$ plaque formation, $A\beta$ 1–42 produced by the amyloid precursor protein undergoes beta and gamma secretase cleavage that can

lead to the formation of monomers, dimers, oligomers, polymers, fibrils, and aggregates.⁷ Among these, the soluble form of $A\beta$ 1–42 oligomers seems to cause the greatest insult, and this process eventually results in neuronal death. This path has been delineated in many reviews.^{8–10} Specifically, $A\beta$ 1–42 oligomers induce several molecules that bring about an increased concentration of glutamate between presynaptic and postsynaptic neurons.¹¹ The resulting inundation of glutamate stimulates not only glutamate receptors on the synapses but also extra-synaptic NMDA NR2B receptors.^{12–14} This causes a cation influx through these receptors that eventually activates a wide range of signaling molecules including MAPKs, CaMKII, and caspases.^{15–19} The downstream signaling of these molecules is well-known to be involved in cell death. Along with these findings, the ERK/JNK/p38, CaMKII, and other pathways have also been shown to be dysregulated in various AD mice models and thus are relevant to the impairment of spatial learning and memory and recognition associated with AD.^{20–22}

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Based on the above information regarding the mechanisms behind AD, many drug candidates from natural resources have been investigated. Notably, *Scutellaria baicalensis*, which is a species of flowering plant from the family Lamiaceae, is the source of baicalein, baicalin, wogonin, norwogonin, oroxylin A, and β -sitosterol, which are major phytochemicals that can be extracted from the plant's root.^{23–25} Each of these compounds, which can be considered to be herbal medicines, has been investigated to a certain degree for its potential pharmaceutical use. We previously screened several herbs and found that baicalein is able to effectively alleviate A β 1–42 oligomer-induced abnormal depolarization of mouse primary cortical glutamatergic neurons based on the results from our screening platform.²⁶ To extend our research in order to find more clinically useful compounds, we screened various derivatives of baicalein and found that homoplantagin is one of the most effective compounds based on our assay. In this study, our aim was to examine the effects of homoplantagin, a major flavonoid glycoside from *Salvia plebeia* R. Br., on animal/neuronal function and then to explore its mechanisms.

MATERIALS AND METHODS

Reagents and Antibodies. The reagents used in this study included Neurobasal-A medium, B27 supplement, L-glutamine (Invitrogen/Gibco, Grand Island, NY, USA), 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2H-tetrazolium bromide (MTT), homoplantagin (catalogue SML2157, $\geq 98\%$ (HPLC), Sigma, St. Louis, MO, USA), bis(1,3-dibutylbarbituric acid)trimethine oxonol [DiBAC4(3)] dye (Life Technologies, Eugene, OR, U.S.A.), and β -amyloid 1–42 (AnaSpec, Fremont, CA, USA). The antibodies used in this study consisted of vGLUT1/2 (SYSY, Gottingen, Germany), 6E10 (Covance, Princeton, NJ, USA), A11 (Invitrogen/Gibco, Grand Island, NY, USA), ERKs/MAPK p42/44 Thr202/Tyr204, ERKs/MAPK p42/44, JNK Thr183/Tyr185, p38 Thr180/Tyr182, and p38 (Cell Signaling Technology, Danvers, MA, USA), JNK, GAPDH, and Type III β -tubulin (abcam, Cambridge, UK). In addition, horseradish peroxidase (HRP) conjugated goat antimouse secondary antibody and goat HRP conjugated antirabbit secondary antibody (Jackson ImmunoResearch, West Grove, PA, USA) were also used. All other chemicals were purchased from Sigma (St. Louis, MO, USA) unless indicated otherwise.

Experimental Animals. All animal experiments were performed with the approval of the Institutional Animal Care and Use Committee of the National Taiwan Normal University (NTNU permission no. 108038). C57BL/6 mice were purchased from the Academia Sinica Animal Center (Taipei, Taiwan). Amyloid precursor protein (APP) transgenic mice (line J20) were used as the AD mouse model; these mice carry the human APP gene with Swedish (K670N/M671L) and Indiana (V717F) familial mutations. After mating, weaning, and genotyping, littermates without transgenic mutants were used as the control and wild-type mice. All mice were housed in a light:dark cycle of 12 h light and 12 h dark and provided *ad libitum* with food and water.

Mouse Primary Cortical/Glutamatergic Neurons. Newborn mice on postnatal days 1 to 3 (P1 to P3) were sacrificed to obtain their cerebral cortex. The protocol has been described previously.^{26,27} In brief, cortex tissue was digested with papain, and the isolated cells were separated by centrifugation. The cells forming the supernatant were collected and delicately laid onto a prepared OptiPrep gradient. After centrifugation of the gradient, cells were collected from the lower layer. These neuronal cells were cultivated on culture dishes previously coated with poly-D-lysine. Immunocytochemistry involving vGLUT1/2 and Type III β -tubule antibodies was used to verify that around 90% of the cellular population was glutamatergic neurons.

A β 42 Oligomers. The preparation of A β 42 oligomers has been described previously.²⁶ Their presence was verified using a dot blot procedure and 6E10/A11 antibodies.

3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium Bromide (MTT) Assays. Cell viability was assessed by the MTT assay as described previously.^{26,27} In brief, the neurons were incubated with the tetrazolium dye MTT. Next, DMSO was used to dissolve the formazan crystals. The absorbance of the resulting solution was measured at 570 nm using a microplate reader.

Measurement of Neuronal Depolarization. The approach used DiBAC4(3), a membrane potential-sensitive fluorescent dye, to measure the changes in neuronal potential; this was developed and established in our laboratory and has been described previously.^{26,27} In brief, the culture medium of the neurons was changed to KRH buffer to which DiBAC4(3) and 4',6-diamidino-2-phenylindole (DAPI) had been added. After incubation, images of neurons were recorded using a fluorescence microscope (Carl Zeiss, Jena, Germany). Using nuclear staining as the reference point for the position of a cell, changes in green fluorescence within the cell were quantified by using MetaMorph software. On average 15 to 25 neurons were quantified from each sample.

Treatment of the Mice and Behavior Testing. The genotyping of newborn mice was performed at 3 weeks of age by extracting genomic DNA from the tails of the mice, and this was followed by PCR using the protocol recommended by the Jackson Laboratory. The mice were then grouped into WT vs. J20 mouse groups. Not until around 5 months old did these groups begin to be injected subcutaneously with mineral oil containing 20% dimethyl sulfoxide (DMSO) with or without homoplantagin to give four groups in total. Specifically, a reference dose of 12 mg/kg body weight homoplantagin was injected 5 days a week for 4 weeks into the two experimental groups at which the dosage was previously published in the literature.²⁸ At around 6 months of age, the four groups of mice were examined using a variety of animal behavior tests.

After the behavior tests, histopathological/hematological analyses were performed; this was at about 7–8 months of age. These procedures have been described previously.^{29,30} In brief, the mice underwent anesthesia, perfusion, and fixation. The kidneys and livers of the mice were removed and postfixed. Assessments of the gross pathology, hematology, and histopathology of the mouse samples were performed by the Mouse Core Facility (Academia Sinica Mice Clinical, Taipei, Taiwan).

Barnes Maze. The equipment, protocol, and acquisition of parameters have been described previously.²⁹ In brief, the mice underwent habituation, the pretraining phase, and the training phase. After these phases, the escape hole/box was removed. The mice were then tested to collect the appropriate measurement using a 3 min probe trial. These measurements include the time lingering in the quadrant used to place the escape hole/box, the average speed of each mouse, and the travel distance of each mouse. Data were analyzed using EthoVision software (XT 17, Noldus, Wageningen, Netherlands).

Novel Object Recognition (NOR) Test. This test was adopted with some modifications from the research of Leger et al.³¹ and Flores et al.³² Briefly, each mouse was habituated to an open field box (45 $\times$ 45 $\times$ 25 cm) and then allowed to explore freely two identical objects for 10 min. After 2 h, the mice were placed back into the same box for 10 min, where one of the objects had been replaced with a novel object, the shape, color, and appearance of which were different. The cumulative duration of exploring both objects over a total time of 20 s was analyzed by using a video tracking system and EthoVision software (XT 17, Noldus, Wageningen, Netherlands).

Morris Water Maze. The protocol has been described previously.³⁰ Briefly, training, testing, and probe trials of the mice were performed in a 120 cm diameter water pool. The distance and trace times were recorded by using a video tracking system. The data were analyzed using EthoVision software (XT 17, Noldus, Wageningen, The Netherlands). Importantly, if the mice drifted around instead of swimming in the water during measurement, the results were omitted; this means that the sample sizes of some groups have been reduced by one or two.

Measurement of Glutamic Oxaloacetic Transaminase (GOT) and Glutamic Pyruvic Transaminase (GPT). Submandibular

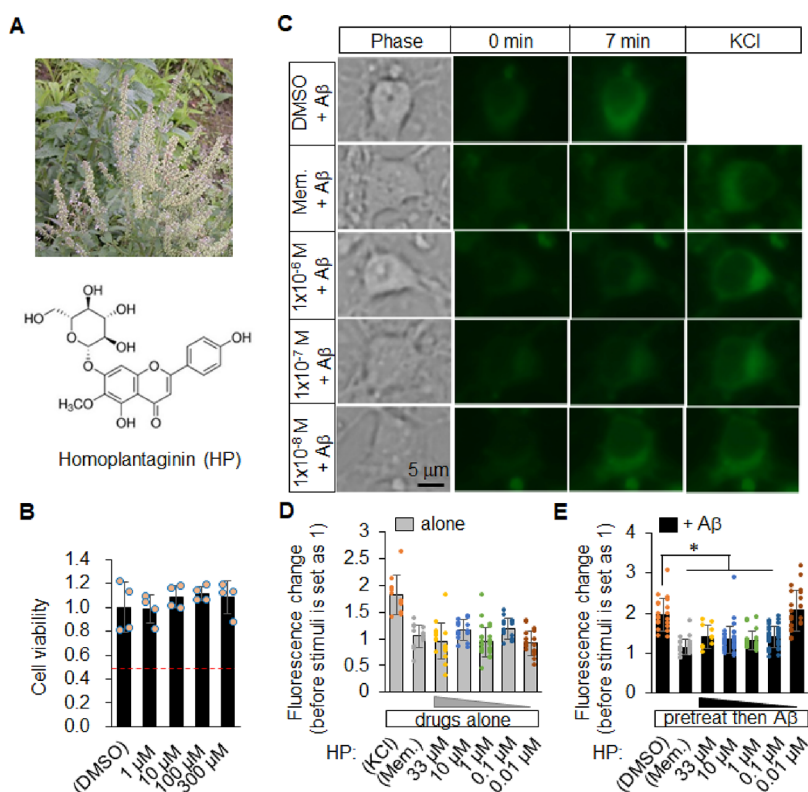


Figure 1. Homoplantagin blocks A β oligomer-induced depolarization. (A) Structure of homoplantagin (HP). (B) Effect of homoplantagin on the viability of primary cortical glutamatergic neurons. (C) The primary cortical glutamatergic neurons were cultivated for 4 days *in vitro* and then preincubated with homoplantagin at different concentrations, as indicated. During the 2 h preincubation, the medium was changed to KRH solution and then adjusted to contain 0.8 μ M DiBAC4(3) for the last 0.5 h. The intensity of the fluorescent changes was recorded after 2.7 μ M A β oligomers stimulation. Memantine 100 μ M was included as a positive control. At the end of each experiment, KCl 45 mM was added to assess the responsiveness of the neurons. (D, E) Images were quantified and analyzed using MetaMorph software. The changes in intensity across the different treatments such as homoplantagin alone and homoplantagin preincubation are shown. [$F = 30.401$, $p < 0.001$, one-way ANOVA, $n = 12-20$]. *, $p < 0.05$ vs A β . HP, homoplantagin; Mem., memantine.

blood was collected from the mice. After centrifugation, the serum samples were transported to the Mouse Core Facility where they used a FUJI DRI-CHEM Clinical Chemistry Analyzer FDC 3500 to measure the GOT/GPT of the samples along with the appropriate standards. Some GOT/GPT data points are absent because the level of hemolysis within samples was high, which made these samples unreliable.

Liver/Kidney Hematoxylin-Eosin (H&E) Staining. Mice under anesthesia underwent surgery in which they were transcardially perfused with 15 mL of 4% paraformaldehyde wash for 5 min. After perfusion, the liver and kidney were collected and trimmed appropriately. These tissues were fixed in 4% paraformaldehyde by shaking for 1 day. The samples were then transported to the Mouse Core Facility, Taiwan. Next, the tissues were dehydrated in serial concentrations of ethanol and embedded in wax. After sectioning by microtome to give 5 μ m thickness samples, these slides were stained in Mayer's hematoxylin for 5 min, rinsed in PBS, and then stained with eosin for 10 s. After dehydration in ethanol and xylene, the slides/images were mounted and recorded using a microscope (Carl Zeiss, Jena, Germany).

Western Blot Analysis. This was carried out as described previously.^{26,27} In brief, the lysate of neuronal cells was separated by SDS-PAGE and then transferred to a PVDF membrane. The membrane was blotted with appropriate primary antibodies and relevant secondary antibodies conjugated with HRP. The bands were visualized and recorded using a CCD camera (ImageQuant LAS 4000, GE Healthcare, Chicago, IL, USA) after treatment with a chemiluminescence kit (Millipore Corporation, Billerica, MA, USA).

Calcium Influx of Glutamatergic Neurons. The protocol has been described previously.³⁰ In brief, the cultivated neurons were

loaded with Fura-2/AM in a culture medium. After being washed, the medium was changed to KRH buffer. The calcium influx was initiated by the various stimuli and recorded every 10 s for at least 15 min; this was carried out using an inverted fluorescent microscopy with a xenon lamp (Leica, Wetzlar, Germany). A23187, a calcium ionophore, was added at the end of the experiment to verify the cellular response. The F340/F380 ratio of the emitted fluorescence after excitation at 340 and 380 nm was analyzed using MetaFluor software (Molecular Devices, San Jose, CA, USA).

Statistical Analysis. The Statistical Package for the Social Sciences (SPSS) software package was used to execute various statistical analyses. One-way or two-way analysis of variance (ANOVA) was used to assess the differences among multiple means. As appropriate, Dunnett's *post hoc* test was adopted to analyze the results further. Some effects of the treatments were evaluated using the two-tailed Student's *t* test. Data are presented as means $\pm$ SD. A *p* value of <0.05 was considered to be statistically significant.

RESULTS

The structure of homoplantagin is shown in Figure 1A. Since there is no neuronal death when the experimental maximum dose has reached 300 μ M in the MTT assay, the half-maximal inhibitory concentration (IC₅₀) of homoplantagin on neuronal survival was not determined (Figure 1B). Despite this, we adopted concentrations well below 300 μ M when performing the relevant experiments. As shown in Figure 1C, treatment with A β oligomers increases fluorescence intensity, and this increase is able to be alleviated by preincubation with

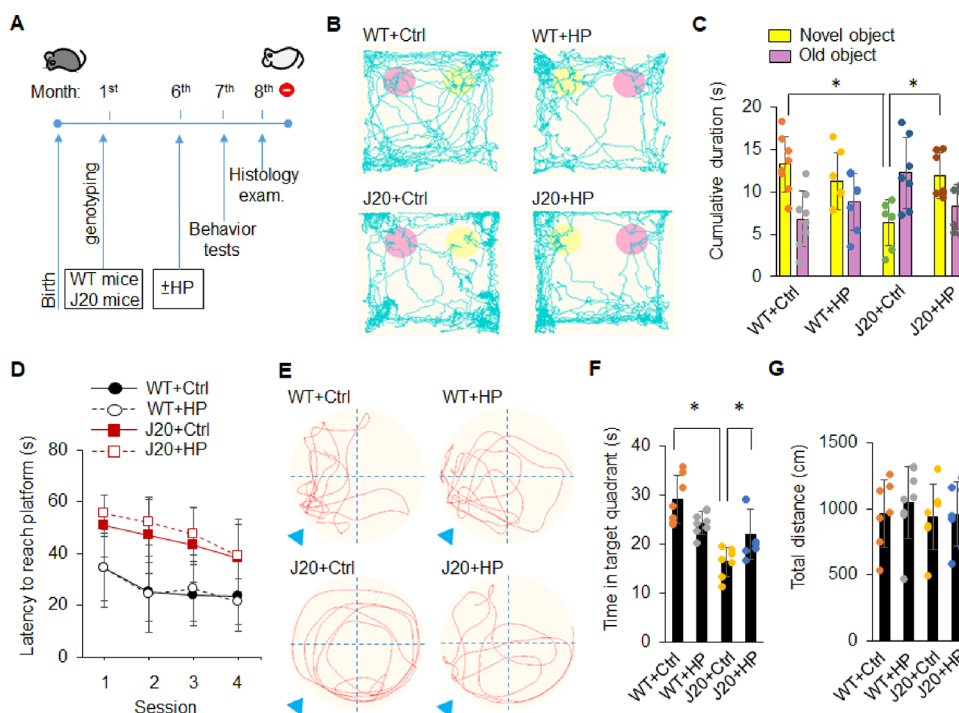


Figure 2. Homoplantagin mitigates the impairment of recognition and the spatial memory presenting in J20 mice. (A) Flowchart of the experimental design and study period. (B) Two h after pre-exposure to two identical objects for 10 min, mice were placed back into the chamber with one novel and one familiar object for 10 min again. A set of representative traces is shown. The yellow area indicates the radius of 3 cm surrounding the novel object, while a similar pink area indicates the familiar one. (C) The cumulative duration added up to 20 s within 10 min for contact with both objects, and this was quantified and plotted. [$F(1,25) = 9.370$, $p = 0.005$, two-way ANOVA, $n = 8$ WT Ctrl, 7 WT HP, 7 J20 Ctrl, and 7 J20 HP]. (D) During the training session for the Morris water maze test, the time needed to reach the hidden platform under water for each mouse was measured. This was done one session per day for 4 days. (E) 48 h after the final training, the probe test was carried out in which the hidden platform was removed. A trace showing how each mouse searched for the hidden platform over 1 min was recorded. A set of representative traces is shown. The color marks indicate the quadrant where the platform used to be located. (F) The time in which each mouse searched for the hidden platform over 1 min was analyzed. [$F(1,25) = 12.231$, $p = 0.002$, two-way ANOVA, $n = 7$ WT Ctrl, 8 WT HP, 7 J20 Ctrl, 7 J20 HP]. (G) The total distance moved in the probe test is shown. WT, wild type; Ctrl, control; HP, homoplantagin.

homoplantagin at 1×10^{-7} M, but not at 1×10^{-8} M. The inhibitory effect of memantine using the same platform has been included as a control. The quantification of the fluorescent intensity using homoplantagin alone (Figure 1D) and preincubation followed by stimulation of A β oligomers (Figure 1E) are presented. Homoplantagin alone has no effect on the membrane potential of cortical glutamatergic neurons. A β oligomers significantly increase fluorescent intensity by around 2-fold and this change can be prevented by preincubation with homoplantagin at concentrations as low as 1×10^{-7} M.

Next, we examined the effect of homoplantagin on the spatial learning and memory behavior using an AD mouse model J20 mice. The Barnes maze is used to test spontaneous learning and memory behavior in mice. As shown in Figure S1, there is no difference in the probe results among the four groups. According to the literature, there have been no reports where the Barnes maze has been used with the J20 mice.³³ Accordingly, we next adopted the novel object recognition (NOR) approach, which is a suitable test for examining AD using the J20 mice. The timelines of behavior tests are shown in Figure 2A. A representative trace is shown in Figure 2B; in this trace, J20 mice that had received the vehicle spent less time exploring the new object compared to wild-type mice that received the vehicle. The change in the pattern of recognition behavior in the J20 control mice was reversed when the J20 mice received homoplantagin treatment. When a cumulative

time of 20 s was used to quantify the duration spent exploring the new object, the vehicle-treated J20 mice were found to spend less than 10 s compared to the homoplantagin-treated J20 mice and wild-type mice, both of which spent more than 10 s (Figure 2C).

Finally, the Morris water maze was adopted as a learning/memory test. During training, there was only a trend toward the latency when reaching the hidden platform to be faster for the wild-type mice than the J20 mice at 7 months old (session 1–4 in Figure 2D). In general, there was no significant difference in this parameter among the four groups after four sessions of training. The probe was performed 48 h after the training and involved the hidden platform being removed. The time spent in the target quadrant by the vehicle-treated wild-type mice group was found to be longer than that of the vehicle-treated J20 mice group. The homoplantagin-treated J20 mice also spent more time searching for the platform in the target quadrant compared to the vehicle-treated J20 control group (Figure 2E). The timings were quantified and found to be statistically significant (Figure 2F). The distance moved within 1 min by the various mice showed no significant differences among the four groups. This indicates that the swimming ability/speed of the four groups was not different (Figure 2G). Thus, homoplantagin is able to improve the impaired recognition and spatial memory behaviors of J20 mice.

While the effects of homoplantagin on memory-relevant parameters were tested, the fundamental safety of homoplantagin was also examined. We simply examined the effects of homoplantagin on inflammation and toxicity detected in blood, liver, and kidney. As shown in Figure S2A, the body weight during the 1 month injection period showed no difference across the four groups. The histology of the liver (Figure S2B) and kidney tissues (Figure S2C) also showed no obvious abnormalities across the four groups. The general indicators glutamic oxaloacetic transaminase (GOT) (Figure S2D) and glutamic pyruvic transaminase (GPT) (Figure S2E) in the blood, which are relevant to the presence of liver inflammation, were also shown to have no difference across the four groups.

Next, we concentrated on revealing the mechanism by which homoplantagin acts at the neuronal level. Homoplantagin is able to effectively inhibit $A\beta$ oligomer-induced abnormal depolarization. This leaves open the question of whether homoplantagin is able to prevent other abnormalities brought about by $A\beta$ oligomers. Thus, we studied the effect of $A\beta$ oligomers on various signaling molecules that have been previously identified in the literature. We found that homoplantagin is able to counteract the effect of $A\beta$ oligomers on ERK, but not on JNK or p38 (Figure 3A). Memantine was used in the same experiment as the positive control. In parallel, C2C12 myoblasts treated with or without UV irradiation were included in the same electrophoretic analysis to demonstrate the specificity of the antibodies and the correct positions of the various signaling molecules. After quantitation, the results show that homoplantagin is able to significantly block $A\beta$ oligomer-induced effects (Figure 3B).

On the basis of the current $A\beta$ hypothesis, $A\beta$ oligomers result in the accumulation of glutamate within the synaptic cleft and this then overactivates NMDA NR2B in the extra-synapse area to bring about neurotoxicity.¹¹ In order to dissect the mechanism behind homoplantagin, we stimulated primary neurons using NMDA to identify a situation that mimics the effects of the $A\beta$ oligomers. As shown in Figure S3, we carried out a series of time course and dose-response experiments involving NMDA treatment on primary neurons. We eventually decided to use 300 μ M NMDA for 3 h as an appropriate parameter that could mimic the abnormal signaling brought about by $A\beta$ oligomers. Using this approach, we found that NMDA induced an increased ERK phosphorylation in primary neurons and this is prevented by 100 μ M homoplantagin (Figure 3C). The phosphorylation of JNK and p38 was not changed after NMDA treatment. Both homoplantagin and MK801 alone have no effect on these molecules. Whether or not the chosen antibodies are able to detect the correct molecules was verified using lysate from C2C12 myoblasts and UV irradiation as shown in lanes 7–9. Quantification of the above blots from three independent experiments is presented in Figure 3D. These findings suggest that the increased ERK phosphorylation brought about by $A\beta$ oligomers and NMDA is able to be significantly blunted by homoplantagin.

$A\beta$ oligomers-induced abnormal depolarization is mediated via a number of different receptors, including AMPAR and NMDAR.²⁶ Therefore, we utilized the agonists AMPA and NMDA in order to reveal the possible molecular mechanism by which homoplantagin acts. First, we preincubated neurons with various doses of homoplantagin ranging from 1×10^{-8} M to 1×10^{-5} M for 2 h and this was followed by AMPA

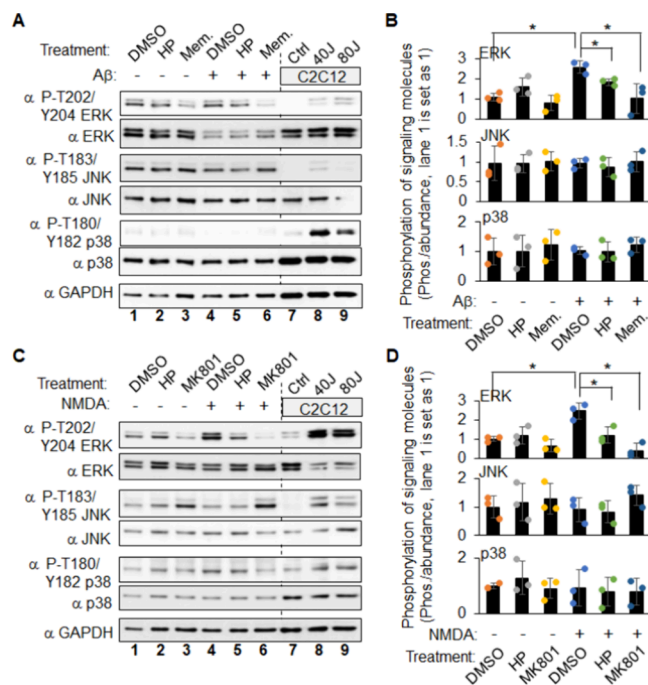


Figure 3. Homoplantagin blunts the increase in ERK phosphorylation brought about by $A\beta$ oligomers and NMDA. (A, C) Primary cortical glutamatergic neurons were preincubated with vehicle, 100 μ M homoplantagin, or the positive control 100 μ M memantine/100 μ M MK801 for 2 h. These neurons were then grown for 9 h with 2 μ M $A\beta$ or 3 h with 300 μ M NMDA. The signaling molecules as indicated were analyzed by Western blotting. The sizes and positions of the various signaling molecules were confirmed by using a lysate from C2C12 myoblasts irradiated through UV light. The blots are representative of an experiment that was repeated three times. (B, D) The various signaling molecules in the blots were quantified using NIH ImageJ software. The y-axis represents the ratio related to the control (lane 1) which was set to 1. [$A\beta$ treatment, $F = 29.271$, $p = 0.001$, one-way ANOVA, $n = 3$; NMDA treatment, $F = 10.892$, $p < 0.001$, one-way ANOVA, $n = 3$]. Ctrl, control; Mem., memantine; HP, homoplantagin.

treatment. Since homoplantagin is a baicalein derivative, we compared the effects of the two compounds. As shown in Figure 4A, homoplantagin across all dosages was able to block AMPA-induced depolarization, while baicalein was not able to do this at lower doses such as 1×10^{-8} and 3×10^{-8} M. Only the images from 3×10^{-8} and 1×10^{-7} M treatments with the two compounds are presented. In the same experiment, an antagonist of AMPAR, CNQX, is included as a positive control. The addition of KCl at the end of the experiment was used to verify the responsiveness of the primary neurons after treatment. The change in fluorescent intensity was quantified and is presented in Figure 4B. AMPA stimulation after vehicle pretreatment shows an increase in fluorescence while pretreatment with homoplantagin across a range of 1×10^{-8} to 1×10^{-5} M was able to significantly inhibit this effect. Notwithstanding the above findings, baicalein acts in the same way but only at the highest doses which ranged from 1×10^{-7} to 1×10^{-5} M. Thus, the blockade of AMPA-induced depolarization brought about by homoplantagin occurs at a much lower concentration than when baicalein is used as the treatment.

Next, we examined the crucial canonical pathway of NMDAR. NMDA-induced depolarization of cortical neurons

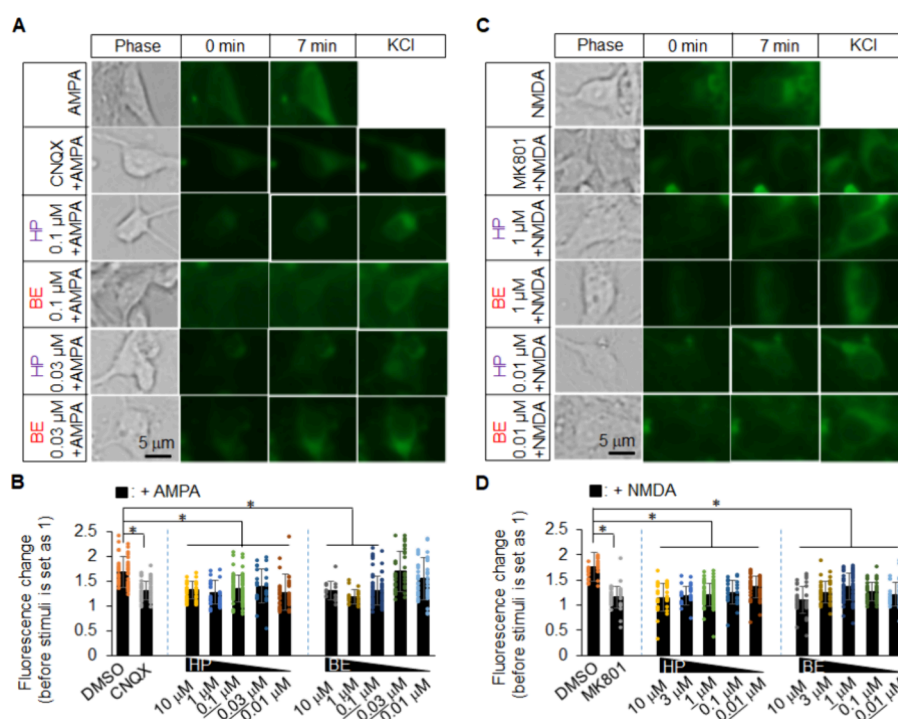


Figure 4. Both homoplantagin and baicalein alleviate AMPA-induced and NMDA-induced depolarization, but homoplantagin is able to counteract AMPA-induced depolarization at a lower concentration than baicalein. (A, C) Primary cortical glutamatergic neurons were preincubated with different concentrations of homoplantagin and baicalein for 2 h. Neuronal depolarization was initiated by the addition of AMPA 20 μ M or NMDA 40 μ M. The responsiveness of the neurons was assessed by adding KCl 45 mM at the end of the experiment. Positive controls include CNQX and MK801, which are respectively inhibitors of AMPAR and NMDAR. (B, D) Images were analyzed using MetaMorph software. The intensity changes at different concentration of homoplantagin and baicalein were quantified. [AMPA treatment, $F = 16.638$, $p < 0.001$, one-way ANOVA, $n = 27$ –50; NMDA treatment, $F = 37.852$, $p < 0.001$, one-way ANOVA, and $n = 32$ –50]. * $p < 0.05$ vs DMSO + AMPA or DMSO + NMDA. HP, homoplantagin; BE, baicalein.

was found to be inhibited by preincubation with a range of homoplantagin doses from 1×10^{-8} to 1×10^{-5} M for 2 h (Figure 4C). In the same experiment, an antagonist of NMDAR, MK801, was included as a positive control. Baicalein was also used across the same range of doses in order to do the comparison. It was found that both homoplantagin and baicalein have similar effects at the above dosages. The fluorescence intensities were quantified and are presented in Figure 4D. The NMDA-induced increase in fluorescence is significantly inhibited by pretreatment with either homoplantagin or baicalein over a range of concentrations. The lowest effective concentration was determined to be 1×10^{-11} M (Figure S4). These findings also indicate that homoplantagin and baicalein both act via AMPAR and NMDAR, possibly directly, to ameliorate neuronal pathophysiology.

The above findings pinpoint the possibility that the direct targets of homoplantagin might be one or more relevant receptors. Since NMDARs themselves are calcium channels, we utilized Fura-2/AM, an indicator of intracellular calcium concentration, to further investigate the action of homoplantagin. First, we found that there is a dose-dependent pattern of increased intracellular calcium associated with the presence of A β oligomers (Figure 5A). After the stimulation with A β oligomers at doses of 0.13 μ M, 0.27 μ M, and 0.54 μ M, there were different levels of gradual increase in the intracellular calcium concentrations within cells. A calcium ionophore, A23187, was added at the end of each experiment in order to verify the cellular response. Preincubation with homoplantagin mitigates the 0.54 μ M A β oligomer-induced

increase in intracellular calcium (Figure 5B). Memantine was included in the same experiments as a positive control. Second, we found that NMDA at doses of 1 μ M, 10–50 μ M, and 150–250 μ M was able to induce gradual/slight, quick/sustained, and quick/biphasic increases in intracellular calcium, respectively (Figure 5C). The observation of a quick/biphasic change brought about by using a high dosage of NMDA is reminiscent of lymphocytes being stimulated by ligands in order to quickly release their intracellular pool of calcium, which is then followed by a sustained extracellular influx. Thus, we used BAPTA/AM and EGTA, which are respectively intracellular and extracellular calcium chelators, to dissect the source of the increase in calcium concentration. As shown in Figure 5D, the phenomenon of calcium biphasic change initiated by 200 μ M NMDA is completely from outside the cell because preincubation of neurons with BAPTA/AM has no effect on the NMDA-induced biphasic pattern. While using calcium-free plus EGTA solution in the outside KRH buffer, the biphasic increase of calcium via NMDA induction is completely blocked. Third, the effect of homoplantagin on the NMDA-induced calcium influx was examined. Homoplantagin alone at a range of doses from 1×10^{-8} to 1×10^{-4} M has no effect on the change in intracellular calcium (Figure 5E). Neurons preincubated with homoplantagin showed an alleviation of the NMDA-induced increase in calcium influx (Figure 5F). Finally, we added homoplantagin during the NMDA-induced steady state of calcium influx to explore possible mechanisms. Interestingly, the addition of homoplantagin immediately inhibits the NMDA-induced calcium

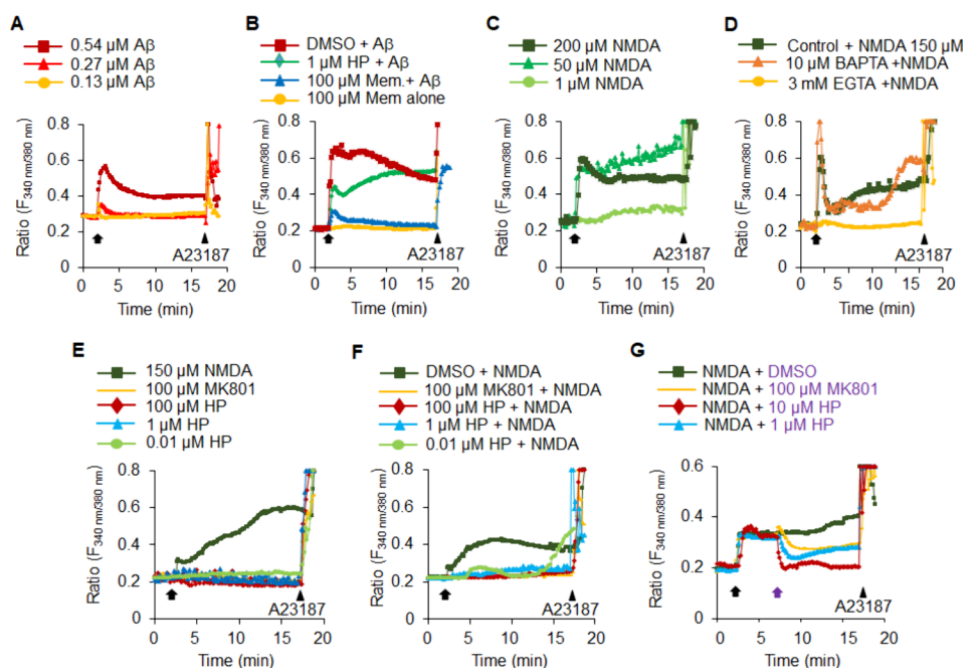


Figure 5. Homoplantagin functions by targeting NMDAR to inhibit $A\beta$ oligomer-induced and NMDA-induced calcium influxes. (A) $A\beta$ oligomers induce a dose responsive increase in intracellular calcium concentration within primary cortical glutamatergic neurons as measured by the Fura-2 F340/F380 ratio of the emitted fluorescent intensity. (B) Homoplantagin blocks the $0.54 \mu\text{M}$ $A\beta$ oligomer-induced change in calcium concentration. Memantine was used as a positive control. (C) NMDA induces a dose-dependent pattern of increased intracellular calcium in primary cortical glutamatergic neurons. (D) Two sets of conditions used for the neuronal incubation. One is that the neurons were incubated with $10 \mu\text{M}$ BAPTA/AM, an intracellular calcium chelator, for 30 min followed by incubation with $1 \mu\text{M}$ Fura-2/AM. The other is that the neurons were incubated with a modified KRH buffer without CaCl_2 in the presence of 3 mM EGTA in the extracellular solution. NMDA $150 \mu\text{M}$ was added at the second min to initiate the neuronal response. (E) Different concentrations of homoplantagin were used as a stimulus to examine their effects on the calcium influx. MK801 $100 \mu\text{M}$ and NMDA $150 \mu\text{M}$ were respectively used as negative and positive controls in the same assay. (F) Preincubation of homoplantagin blocks the NMDA-induced calcium influx. MK801, a NMDAR inhibitor, was used as a positive control. (G) Different concentrations of homoplantagin were added at the 5th min during the plateau phase of the NMDA-induced calcium influx. DMSO and MK801 $100 \mu\text{M}$ were respectively used as negative and positive controls. The arrow at the second min indicates the addition of the stimulus, while the arrow at 17th min indicates the addition of A23187, a calcium ionophore. Number of neurons = 15–20. Mem., memantine; HP, homoplantagin.

influx in a dose-dependent pattern, as shown in Figure 5G. DMSO and MK801 were included in this experiment as controls. Taken together, these results show that homoplantagin seems to be able to act directly onto the NMDAR molecule.

DISCUSSION

Utilizing a drug screening platform,²⁶ a pure compound, homoplantagin (6-methoxy-4',5-dihydroxyflavone-7-O-glucoside), was identified as being able to effectively counteract $A\beta$ oligomers-induced abnormal depolarization. Homoplantagin is a pure natural flavonoid glycoside that can be extracted from plants, such as *Salvia plebeia* R. Br. We found that homoplantagin is able to prevent several pathophysiological states caused by $A\beta$ oligomers in primary glutamatergic neurons.

Homoplantagin has been reported to be beneficial for cells when they are insulted by a variety of adverse stimuli. Homoplantagin is able to ameliorate high-glucose-induced apoptosis in vascular endothelial cells. This effect involves the AMPK/mTOR pathway and an inhibition of the store-operated calcium entry channel.^{34,35} The inflammation of vascular endothelial cells caused by palmitic acid is attenuated by the homoplantagin treatment. This mechanism would seem to be mediated via blocking of the toll-like receptor-4 and NLRP3 pathways.³⁶ In a mice model of hepatic injury induced

by an injection of a *Bacillus Calmette-Guérin* (BCG)/lipopolysaccharide (LPS) mixture, homoplantagin is able to reverse a variety of abnormal inflammation indices including tumor necrosis factor- α (TNF- α), interleukin-1 (IL-1), alanine aminotransferase (ALT), and aspartate aminotransferase (AST).³⁷ Here we are for the first time reporting that homoplantagin is able to alleviate $A\beta$ oligomer-initiated dysfunction in both primary neurons and an AD animal model. We have also pinpointed that homoplantagin would seem to directly target the NMDAR protein. It is worth noting that we have not directly measured whether homoplantagin is able to pass through blood-brain-barrier (BBB). However, homoplantagin has been predicted to be able to diffuse through the BBB by the “parallel artificial membrane permeability assay” which mimics the blood–brain barrier (PAMPA-BBB).³⁸ Moreover, the promising results in our animal behavior test indicate that homoplantagin and/or its metabolites most likely are capable of crossing the BBB and, in doing so, remain functional.

Homoplantagin consists of a flavone plus glucose. Homoplantagin can also be viewed as a link of hispidulin to UDP-glucose. Actually, four flavonoids, homoplantagin, hispidulin, nepetin, and nepetin-7-glucoside, can be extracted from the Asian herb *Salviae plebeiae*.³⁹ Hydroxyl derivatives of flavones include a number of well-studied compounds including various catechins, epigallocatechin gallate (EGCG),

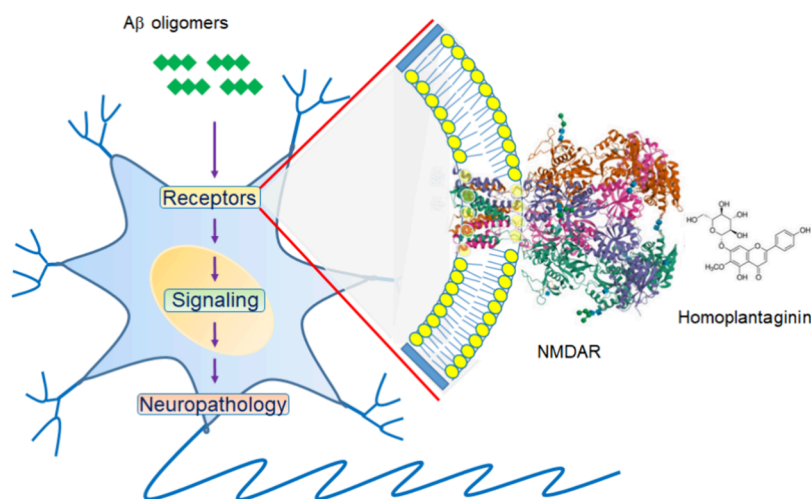


Figure 6. Proposed model for homoplantagin acting on NMDAR. The NMDAR structure is adopted from RCSB Protein Data Bank (RCSB PDB) (<https://www.rcsb.org/>) using ID: 7SAD. Homoplantagin might directly associate with the NMDAR itself, bringing about competitive, noncompetitive, or even uncompetitive binding, and thus inhibiting the functioning of the relevant receptors. The detailed mechanism needs further investigation.

and baicalein. All of these components are well-known to act as anti-inflammatory and antioxidative compounds. Catechins are known to be beneficial because they have antioxidative effects and can be active against pathogens.⁴⁰ In addition to EGCG's effects on immunity, EGCG has recently been targeted as a potential anticancer therapeutic agent.⁴¹ We previously found that baicalein is able to effectively block the Aβ oligomer-induced pathophysiological state, including abnormal depolarization.²⁶ In this study, homoplantagin was found to have a similar effect, and the action of this drug was further investigated. On the basis of homoplantagin's structure, a variety of derivatives evolved from this flavone ought to provide a great resource that can be used for therapeutic drug development, in particular for targeting AD.

Several signaling molecules, including ERK, p38, JNK, and CaMKII, have been found to be affected in neurons treated with Aβ or in Aβ related animal models.^{11,42} However, we found that only an increase in ERK phosphorylation was detected when the above signaling proteins were examined. These discrepancies might be attributed to differences in the neuronal models used compared to those in the present investigation. Some pathways associated with Aβ-induced neurotoxicity have been identified using studies based on neuronal cell lines, these include human-derived neurotypic SH-SY5Y cells⁴³ and human neuroblastoma cell line SK-N-SH cells.⁴⁴ Other studies have used primary hippocampus neuronal cells.^{45,46} However, more complicated signaling modifications, such as dysregulation of calcium and potassium ion channels, have also been revealed using brain slices or brain tissue.^{47,48} In the current study, the high purity of the primary glutamatergic neurons probably reflects better cell physiology compared to other cell lines, but this may have resulted in less complicated processes being detected than brain tissues that contain a mixture of cell types. The variation in properties among these target materials is one of the rationales for us spending effort on pinpointing the optimal time course/dose response for NMDA before examining the effects of homoplantagin on neurons (Figure S3). On the other hand, ERK1/2 activation has been shown to be an essential aspect of memory and learning. Nevertheless, ERK1/2 overactivation is known to cause ROS accumulation that can

then result in oxidative stress.^{16,49} In this study, the phenomena of Aβ-induced and NMDA-induced increases in ERK phosphorylation can be specifically prevented by treatment with homoplantagin or memantine. These findings suggest that small molecules that are capable of targeting the Aβ-induced ERK pathway might also be merit/potential drug candidates for treating AD.

The primary glutamatergic neurons from the cortex used in our study provide an excellent *ex vivo* model for investigating the effects of homoplantagin on Aβ oligomer-induced neuropathology. The experimental outcome is able to be much more precisely interpreted in terms of cause/effect at the cellular level. However, one of the limitations of the present study is that the complex nature of the interaction between neurons and non-neurons and/or among different brain cells/tissues *in vivo* can not be reflected in such a relatively simple cellular model. Moreover, tau-induced neuropathy of neurons is another critical marker when studying AD; this is where hyperphosphorylation of tau occurs within the neurons. Whether homoplantagin can affect the neuropathology brought about by hyperphosphorylation of tau is another unanswered question and thus is a limitation of the study. To explore this, there is a need to perform gene modification of primary glutamatergic neurons. Thus, the effects of homoplantagin on brain tissues by exploring the morphological and biochemical aspects and by investigating tau-induced neuropathology need further investigation.

Aβ oligomers have been found to be able to promptly increase intracellular calcium via NMDAR and, to a lower extent, via AMPAR.^{50,51} In our study, this was confirmed using glutamatergic neurons. The system was further simplified using NMDA as a stimulus. Such an NMDA-induced calcium influx was able to be blocked either by homoplantagin preincubation or by homoplantagin addition after ligand stimulation. Thus, we speculate that NMDAR is one of the major targets for homoplantagin. Both homoplantagin and the NMDAR antagonist MK801 produce similar inhibitory effects when added at fifth min in terms of the NMDA-induced calcium influx (Figure 5). This result provides proof-of-concept support for our hypothesis. Furthermore, in a preliminary analysis using molecular docking, some plausible

binding sites on NMDAR for homoplantaginins were able to be identified. However, the precise exploration of these binding sites needs a more comprehensive investigation. In the cartoon presented in Figure 6, we schematically propose the molecular mechanism by which homoplantaginins acts in neurons. Homoplantaginins may directly associate with NMDAR itself through competitive binding, noncompetitive binding, or uncompetitive binding in order to inhibit the functioning of the receptor.

As a natural product, homoplantaginins has been identified to effectively alleviate A β -induced abnormalities. At the organism level, homoplantaginins ameliorates the impairment of recognition and spatial memory behavior in an AD mice model. The increase in ERK phosphorylation triggered by A β oligomers/NMDA can be reversed by homoplantaginins. Furthermore, homoplantaginins reduces the A β oligomer-induced abnormal depolarization, whereby it seems to be mediated via inhibition of the NMDAR/AMPA pathways. Remarkably, NMDA-induced calcium influx during a steady state is abrogated right after the addition of homoplantaginins, which indicates that the compound is likely to directly bind and interfere with the activity of NMDAR. Therefore, homoplantaginins has great potential and merit as a candidate for the development of new therapeutic drugs.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jafc.4c07659>.

Effect of homoplantaginins on mice via the Barnes maze; effect of homoplantaginins on inflammation and morphology of tissues; effects of the NMDA-induced time-course and dose–response on various signaling molecules; minimal effective dose of homoplantaginins on the inhibition of NMDA-induced depolarization; and all full blots shown in Figure 3 (PDF)

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Author Contributions

T.Y.C., Y.R.C., M.L.H., Y.T.L., and C.H.W. designed and performed the experiments, as well as assembled, analyzed, and interpreted the data. C.A.Y. provided essential materials and interpreted the data. W.C.Y. analyzed and interpreted the data. W.L. and Y.L. designed the experiments, interpreted the data, supervised the study, and wrote the manuscript. All the authors have read and approved the manuscript.

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Notes

The authors declare no competing financial interest.

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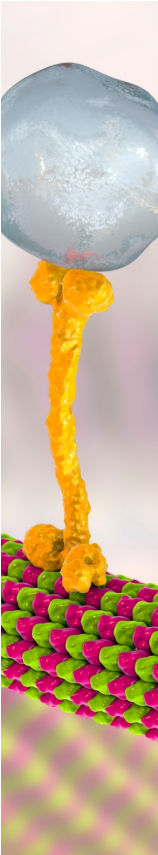
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